Answer Key

Other (142 Questions)

Q1. (D)	Q2. (D)	Q3. (A)	Q4. (A)	Q5. (D)
Q6. (A)	Q7. (C)	Q8. (D)	Q9. (D)	Q10. (D)
Q11. (C)	Q12. (C)	Q13. (A)	Q14. (A)	Q15. (D)
Q16. (C)	Q17. (A)	Q18. (B)	Q19. (A)	Q20. (C)
Q21. (A)	Q22. (B)	Q23. (B)	Q24. (C)	Q25. (A)
Q26. (B)	Q27. (B)	Q28. (C)	Q29. (B)	Q30. (B)
Q31. (B)	Q32. (D)	Q33. (B)	Q34. (A)	Q35. (C)
Q36. (A)	Q37. (A)	Q38. (C)	Q39. (A)	Q40. (A)
Q41. (B)	Q42. (A)	Q43. (D)	Q44. (B)	Q45. (A)
Q46. (D)	Q47. (B)	Q48. (A)	Q49. (A)	Q50. (A)
Q51. (D)	Q52. (B)	Q53. (A)	Q54. (C)	Q55. (D)
Q56. (C)	Q57. (D)	Q58. (D)	Q59. (B)	Q60. (A)
Q61. (A)	Q62. (A)	Q63. (D)	Q64. (A)	Q65. (B)
Q66. (A)	Q67. (D)	Q68. (B)	Q69. (A)	Q70. (D)
Q71. (A)	Q72. (C)	Q73. (D)	Q74. (A)	Q75. (D)
Q76. (D)	Q77. (C)	Q78. (A)	Q79. (D)	Q80. (C)
Q81. (A)	Q82. (D)	Q83. (C)	Q84. (A)	Q85. (A)
Q86. (A)	Q87. (C)	Q88. (C)	Q89. (D)	Q90. (B)
Q91. (B)	Q92. (C)	Q93. (A)	Q94. (B)	Q95. (A)
Q96. (C)	Q97. (B)	Q98. (A)	Q99. (D)	Q100.(B)
Q101.(D)	Q102.(C)	Q103.(C)	Q104.(A)	Q105.(C)

Q106.(A)	**Q107.(D)**	**Q108.(A)**	**Q109.(A)**	**Q110.(A)**
Q111.(D)	**Q112.(C)**	**Q113.(B)**	**Q114.(C)**	**Q115.(D)**
Q116.(B)	**Q117.(B)**	**Q118.(C)**	**Q119.(A)**	**Q120.(B)**
Q121.(A)	**Q122.(B)**	**Q123.(C)**	**Q124.(C)**	**Q125.(B)**
Q126.(D)	**Q127.(C)**	**Q128.(D)**	**Q129.(A)**	**Q130.(A)**
Q131.(A)	**Q132.(C)**	**Q133.(C)**	**Q134.(D)**	**Q135.(A)**
Q136.(D)	**Q137.(A)**	**Q138.(B)**	**Q139.(D)**	**Q140.(B)**
Q141.(C)	**Q142.(A)**			

Solutions

Q1. Solution

Correct Answer: (D)

Boiling point of water, $=$ UFP 65°

Freezing point of water, $=$ LFP -15°

$\frac{\mathrm{X-LFP}}{\mathrm{UFP-LFP}}=\frac{\mathrm{T_F-32}}{212-32}$

$\Rightarrow \frac{-95-(-15)}{65-(-15)}=\frac{\mathrm{T_F-32}}{180}$

$\Rightarrow \frac{-80}{80}=\frac{T_F-32}{180} \Rightarrow T_F=-180+32$

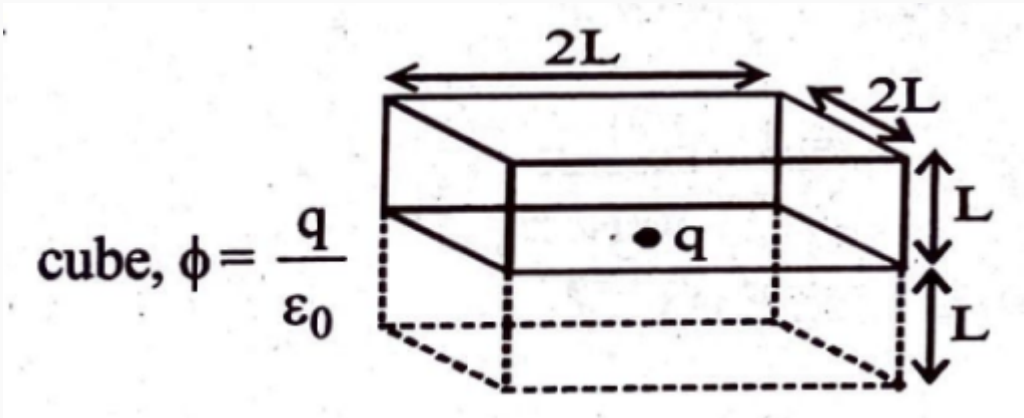
$\Rightarrow \mathrm{T_F}=-148^{\circ}\mathrm{F}$

Q2. Solution

Correct Answer: (D)

After placing similar cubic at the bottom, we get cube of side $2\ \mathrm{L}$ having change q at its centre.

From the Gauss's law electric flux through whole



Flux passing through shaded face

$\phi=\frac{\mathrm{q}/\varepsilon_0}{6}=\frac{\mathrm{q}}{6\varepsilon_0}$

Q3. Solution

Correct Answer: (A)

Thermal energy is given by

$\mathrm{H}=\mathrm{P}\times\mathrm{t}=\frac{\mathrm{V}^2}{\mathrm{R}}\times\mathrm{t}$

Here, voltage V is same.

$\therefore \quad \frac{\mathrm{H_1}}{\mathrm{H_2}}=\frac{\frac{\mathrm{V^2t}}{\mathrm{R}}}{\frac{\mathrm{V^2t}}{3\mathrm{R}}}=3:1$

Q4. Solution

Correct Answer: (A)

$$v = \mathrm{r}^{\mathrm{a}} \rho^{\mathrm{b}} \mathrm{s}^{\mathrm{c}} \Rightarrow [v] = [\mathrm{r}]^{\mathrm{a}} [\rho]^{\mathrm{b}} [\mathrm{s}]^{\mathrm{c}}$$

$$\Rightarrow \quad \left[\mathrm{T}^{-1}\right] = \left[\mathrm{L}^{1}\right]^{\mathrm{a}} \left[\mathrm{M}^{1}\, \mathrm{L}^{-3}\right]^{\mathrm{b}} \left[\frac{\mathrm{MLT}}{\mathrm{L}}\right]^{-2}$$

$$\Rightarrow \mathrm{T}^{-1} = \mathrm{M}^{\mathrm{b}+\mathrm{c}} \cdot \mathrm{L}^{\mathrm{a}-3\,\mathrm{b}} \cdot \mathrm{T}^{-2c}$$

$$\Rightarrow \mathrm{b} + \mathrm{c} = 0, \mathrm{a} - 3\,\mathrm{b} = 0, -2\mathrm{c} = -1$$

$$\Rightarrow \quad \mathrm{c} = \frac{1}{2},\ \mathrm{b} = -\frac{1}{2}, \mathrm{a} - 3\,\mathrm{b} = 0;$$

$$\mathrm{a} + \frac{3}{2} = 0 \Rightarrow \mathrm{a} = -\frac{3}{2}$$

Q5. Solution

Correct Answer: (D)

Velocity of train A

$\mathrm{V_A} = 90 \frac{\mathrm{km}}{\mathrm{hr}} = 90 \times \frac{5}{18} = 25\ \mathrm{m/s}$

Velocity of train B

$\mathrm{V_B} = 54 \frac{\mathrm{km}}{\mathrm{hr}} = 54 \times \frac{5}{18} = 15\ \mathrm{m/s}$

Velocity of train B w.r.t. train

$$\begin{aligned} A &= \vec{V}_B - \vec{V}_A \\ &= 15 - (-25)\mathrm{m/s} \\ &= 40\ \mathrm{m/s} \end{aligned}$$

Time of crossing $= \frac{\text{length of train}}{\text{relative velocity}}$

$(8) = \frac{\ell}{40}$

$\ell = 8 \times 40 = 320$ meter.

Q6. Solution

Correct Answer: (A)

Given,

Initial velocity of projectile, $u = 40\ \mathrm{m/s}$

Angle, $\theta = 30^{\circ}$

Time of flight

$\mathrm{T} = \frac{2\mathrm{u}\sin\theta}{\mathrm{g}} = \frac{2 \times 40 \times 1}{10 \times 2} = 45\ \left(\because \mathrm{g} = 10\ \mathrm{m/s^2}\right)$

It means projectile is at maximum height at $t = 2$ s. At maximum height vertical component of velocity is zero.

$\text{Velocity at t} = 2\ \mathrm{s} = \mathrm{V_x} = \mathrm{u}\cos\theta = 40\cos 30^{\circ}$

$= 20\sqrt{3}\ \mathrm{ms^{-1}}.$

Q7. Solution

Correct Answer: (C)

The magnetic moment associated with circular coil,

$M = NIA$

$M_A = M_B$

$\therefore N_A I_A A_A = N_B I_B A_B$

$\therefore N_A I_A \pi (0.1)^2 = N_B I_B \pi (0.2)^2 \left(\because A = \pi r^2\right)$

$\therefore N_A I_A = 4 N_B I_B$

Q8. Solution

Correct Answer: (D)

Induced emf, $\varepsilon = -\mathrm{L}\frac{\mathrm{dI}}{\mathrm{dt}} \Rightarrow 20 = -\mathrm{L}\frac{(0-2)}{10^{-3}}$ Hence, inductance of the coil.

$\mathrm{L} = 10\mathrm{mH}$

Q9. Solution

Correct Answer: (D)

$$\mathrm{K.E.} = \frac{\mathrm{p}^2}{2\,\mathrm{m}}$$

$$\therefore \frac{\mathrm{K}_1}{\mathrm{K}_2} = \frac{\mathrm{p}_1^2}{2\,\mathrm{m}_1} \times \frac{2\,\mathrm{m}_2}{\mathrm{p}_2^2} = \frac{\mathrm{m}_2}{\mathrm{m}_1} = \frac{16}{9}$$

$$\therefore \frac{\mathrm{m}_1}{\mathrm{m}_2} = \frac{9}{16}$$

Q10. Solution

Correct Answer: (D)

$T \cdot E = K \cdot E + P.E$

$\Rightarrow K \cdot E = T \cdot E - K \cdot E$

$= \frac{1}{2}k^2 - \frac{1}{2}k^2$

$\Rightarrow KE = \frac{1}{2}m\omega^2\left(A^2 - x^2\right) \quad \left[\because k = m\omega^2\right]$

KE is maximum at $\mathrm{x} = 0$ \& KE is zero when $\mathrm{x} = \mathrm{A} \therefore \mathrm{K.E}$ vs x graph is parabola.

Q11. Solution

Correct Answer: (C)

Radius of circle, $\mathrm{r}=\mathrm{a}$

Centrifugal force, $F=m\omega^2 r$

$$\Rightarrow \frac{\mathrm{Gmm}}{(2\mathrm{a})^2}=\mathrm{m}\omega^2\mathrm{a}$$

$$\left(\because \mathrm{F}=\frac{\mathrm{GMM}_2}{\mathrm{d}}\right)$$

Here $M_1=m; M_2=m$ and $d=2a$

$\Rightarrow$ angular speed, $\omega=\sqrt{\frac{Gm}{4a^3}}$ ~

Q12. Solution

Correct Answer: (C)

Equivalent resistance in parallel combination is given by

$$\frac{1}{R_{\text{eq}}}=\frac{1}{R_1}+\frac{1}{R_2}$$

$$\Rightarrow R_{\text{eq}}=\frac{R_1R_2}{R_1+R_2}=\frac{10\times 15}{10+15}=6$$

Differentiating both sides, we get

$$\frac{\Delta R_{\text{eq}}}{\mathrm{R}_{\text{eq}}^2}=\frac{\Delta \mathrm{R}_1}{\mathrm{R}_1^2}+\frac{\Delta \mathrm{R}_2}{\mathrm{R}_2^2}$$

$$\Rightarrow \frac{\Delta \mathrm{R}_{\text{eq}}}{\mathrm{R}_{\text{eq}}}=\left(\frac{\Delta \mathrm{R}_1}{\mathrm{R}_1^2}+\frac{\Delta \mathrm{R}_2}{\mathrm{R}_2^2}\right)\mathrm{R}_{\text{eq}}$$

$$=\left(\frac{0.5}{100}+\frac{0.5}{225}\right)\cdot 6=\left(\frac{6\times 0.5}{25}\right)\left(\frac{1}{4}+\frac{1}{9}\right)=\frac{13}{300},$$

$$\frac{\Delta \mathrm{R}_{\text{eq}}}{\mathrm{R}_{\text{eq}}}\times 100=\frac{13}{3}=4.33\%$$

Q13. Solution

Correct Answer: (A)

(a) Given,

Capacitive reactance, $\mathrm{X_C}=100\Omega$

Inductive reactance, $\mathrm{X_L}=200\Omega$

Resistance, $R=100\Omega$

Impedance, $\mathrm{Z}=\sqrt{\mathrm{R}^2+(\mathrm{X_L}-\mathrm{X_C})^2}$

$$Z=\sqrt{100^2+(200-100)^2}=100\sqrt{2}\Omega$$

RMS value of current,

$i_{\text{rms}}=\frac{V_{\text{rms}}}{Z}=\frac{200\sqrt{2}}{100\sqrt{2}}=2\text{ A}$ ~

Q14. Solution

Correct Answer: (A)

For an EM wave

$\overrightarrow{\mathrm{B}}=\frac{1}{\omega}(\overrightarrow{\mathrm{K}} \times \overrightarrow{\mathrm{E}})$,

Q15. Solution

Correct Answer: (D)

Angle of deviation for first prism

$\delta_1=A_1\left(\mu_1-1\right)=6(1.54-1)$

Angle of deviation for second prism

$\delta_2=A_2\left(\mu_2-1\right)$

$=A_2(1.72-1)$

For dispersion without deviation

$\delta_1=\delta_2$

$\Rightarrow 6^{\circ}(1.54-1)=\mathrm{A}_2(1.72-1)$

$\Rightarrow \mathrm{A}_2=\dfrac{6^{\circ} \times 0.54}{0.72}=\dfrac{18^{\circ}}{4}=4.5^{\circ}$.

Q16. Solution

Correct Answer: (C)

Given that torque, $\vec{\tau}=5 \hat{i}+3 \hat{j}-7 \hat{k}$

Position vector, $\overrightarrow{\mathrm{r}}=2 \hat{\mathrm{i}}++2 \hat{\mathrm{j}}+\hat{\mathrm{k}}$

$$\vec{\tau}=\overrightarrow{\mathrm{r}} \times \overrightarrow{\mathrm{F}}=\begin{array}{ccc} \mathrm{i} & \mathrm{j} & \mathrm{k} \\ 2 & 2 & 1 \\ 5 & 3 & -7 \end{array} \wedge$$

$=\hat{\mathrm{i}}(-14-3)-\hat{\mathrm{j}}(-14-5)+\hat{\mathrm{k}}(6-10)$

$=-17 \hat{\mathrm{i}}+19 \hat{\mathrm{j}}-4 \hat{\mathrm{k}}$

Q17. Solution

Correct Answer: (A)

Potential due to sphere

$\mathrm{V}=\frac{\mathrm{GM}}{2 \mathrm{R}^3}\left(3 \mathrm{R}^2-\mathrm{r}^2\right)$

At surface $r=R \Rightarrow V=\left(\frac{G M}{R}\right)$

At centre, $r=0$

$\therefore \mathrm{V}_0=\frac{3 \mathrm{GM}}{2 \mathrm{R}}=\left(\frac{3 \mathrm{V}}{2}\right) \wedge$

Q18. Solution

Correct Answer: (B)

Elongation produced in the wire is given b

$\Delta \mathrm{L}=\frac{\mathrm{FL}}{\mathrm{AY}}$

Here, $L=$ length of wire

$A=$ area of cross-section of wire

$\mathrm{Y}=$ Young's moduli of material of wire.

$\because \mathrm{F}$ and L is same and

$$\frac{\mathrm{Y_A}}{\mathrm{Y_B}}=\frac{1}{4} \quad \text{(given)}$$

$$\frac{\mathrm{A_A}}{\mathrm{A_B}}=\frac{1}{3} \quad \text{(given)} \qquad \wedge$$

$$\therefore \frac{\Delta \mathrm{L_A}}{\Delta \mathrm{L_B}}=\frac{\mathrm{A_B}}{\mathrm{A_A}} \frac{\mathrm{Y_B}}{\mathrm{Y_A}}=\frac{3}{1} \times \frac{4}{1}=12$$

Q19. Solution

Correct Answer: (A)

De-Broglie wavelength,

$$\lambda=\frac{\mathrm{h}}{\mathrm{mv}}=\frac{\mathrm{h}}{\sqrt{2 \mathrm{mK}}}=\frac{\mathrm{h}}{\sqrt{2 \mathrm{mq} \Delta \mathrm{V}}}$$

$$\therefore \frac{\lambda_\alpha}{\lambda_{\mathrm{p}}}=\sqrt{\frac{2\ \mathrm{m_p V_p q_p}}{2\ \mathrm{m}_\alpha \mathrm{V}_\alpha \mathrm{q}_\alpha}}$$

$\because \mathrm{V_p}=2\ \mathrm{V}$ and $\mathrm{V}_\alpha=4\ \mathrm{V}$ (given)

$\mathrm{m}_\alpha=4\ \mathrm{m_p}$ and $\mathrm{q}_\alpha=2 \mathrm{q_p}$

$\therefore \frac{\lambda_\alpha}{\lambda_p}=\sqrt{\frac{1 \times 2 \times 1}{4 \times 4 \times 2}}=\frac{1}{4} \Rightarrow \lambda_p: \lambda_\alpha=4: 1$!

Q20. Solution

Correct Answer: (C)

From the Bohr's quantization rule

Angular momentum $\mathrm{L}=\frac{\mathrm{nh}}{2 \pi}, \mathrm{L}_1=\frac{1 . \mathrm{h}}{2 \pi}=\mathrm{L}$ ($n=1$ for first orbit)

in ($n=2$ for second orbit) $L_2=\frac{2 h}{2 \pi}=2 L$

Hence, change in angular momentum $\mathrm{L}_2-\mathrm{L}_1=2\ \mathrm{L}-\mathrm{L}=\mathrm{L}$,

Q21. Solution

Correct Answer: (A)

In adiabatic process

$\mathrm{PV}^{\gamma}=$ constant

$$\Rightarrow \quad \mathrm{P}_1 \mathrm{V}_1^{\gamma}=\mathrm{P}_2 \mathrm{V}_2^{\gamma}$$

$$\Rightarrow \quad \frac{\mathrm{P}_1}{\mathrm{P}_2}=\left(\frac{\mathrm{V}_2}{\mathrm{V}_1}\right)^{\gamma} \Rightarrow \frac{81}{16}=\left(\frac{27}{8}\right)^{\gamma}$$

$$\Rightarrow\left(\frac{9}{4}\right)^2=\left(\frac{9 \times 3}{4 \times 2}\right)^{\gamma} \Rightarrow\left(\frac{9}{4}\right)^{2-\gamma}=\left(\frac{3}{2}\right)^{\gamma}$$

$$\Rightarrow\left(\frac{9}{4}\right)^{2-\gamma}=\left(\frac{9}{4}\right)^{\gamma / 2} \Rightarrow 2-\gamma=\frac{\gamma}{2}$$

$$\Rightarrow 2=\frac{3}{2} \gamma \Rightarrow \gamma=\frac{4}{3}$$

Q22. Solution

Correct Answer: (B)

Average kinetic energy for diatomic gases

$$\mathrm{K}_{\mathrm{av}}=\frac{5}{2} \mathrm{kT}$$

$$\therefore \frac{\left(\mathrm{K}_{\mathrm{av}}\right)_{\mathrm{H}}}{\left(\mathrm{K}_{\mathrm{av}}\right)_{\mathrm{O}}}=\frac{(27+273)}{(27+273)}=1$$

Q23. Solution

Correct Answer: (B)

Given that number of turns, $n=1200$

Current, $\mathrm{I}=2 \mathrm{~A}$

Magnetic field at centre inside the solenoid is given by,

$\mathrm{B}=\mu_0 \mathrm{nI}$

So magnetic intensity at centre of the solenoid,

$$\mathrm{H}=\frac{\mathrm{B}}{\mu_0}=\mathrm{nI}=\left(\frac{1200}{2}\right)(2) \quad\left(\because \mathrm{B}=\mu_0 \mathrm{H}\right)$$

$$\mathrm{H}=1.2 \times 10^3 \mathrm{Am}^{-1}$$

Q24. Solution

Correct Answer: (C)

It is given that

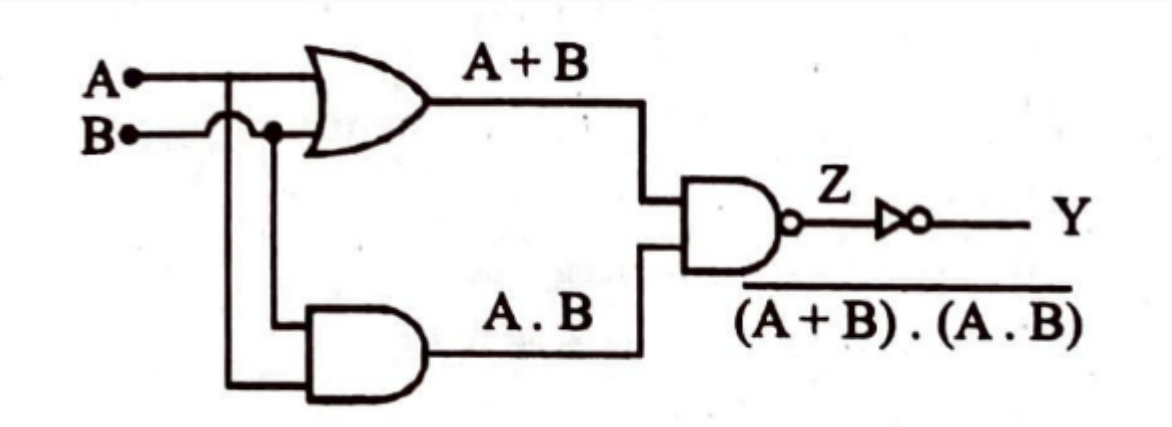
$$\left(\frac{\mathrm{T}_1}{2}\right)_{\mathrm{A}}=\left(\mathrm{T}_{\mathrm{av}}\right)_{\mathrm{B}}$$

$$\because \mathrm{T}_{1/2}=\frac{\ell \mathrm{n} 2}{\lambda} \text{ and } \mathrm{T}_{\mathrm{av}}=\frac{1}{\lambda}$$

$$\therefore \frac{\ell \mathrm{n} 2}{\lambda_{\mathrm{A}}}=\frac{1}{\lambda_{\mathrm{B}}} \Rightarrow \lambda_{\mathrm{A}}=\lambda_{\mathrm{B}} \ell n 2$$

Q25. Solution

Correct Answer: (A)



$Y = \overline{Z} = (A + B) \cdot (A \cdot B) = A \cdot A \cdot B + AB \cdot B$

$Y = AB + AB = A \cdot B$

$\therefore$ It is an AND Gate.

Q26. Solution

Correct Answer: (B)

Let r be the radius of small drops of water.

R = radius of big drop formed as volume remain same.

$\therefore 8.\ \frac{4}{3}\pi r^3 - \frac{4}{3}\pi R^3 \Rightarrow R = 2r$

Terminal velocity,

$$v_T = \frac{2}{9\eta}(\rho - \sigma)r^2\ g$$

$\therefore v_T \propto r^2$

$$\therefore \frac{v_1}{v_2} = \left(\frac{r}{R}\right)^2$$

$$\Rightarrow \frac{10}{v_2} = \left(\frac{1}{2}\right)^2 \ (\because v_1 = 10\ \text{cm/s given})$$

$\Rightarrow v_2 = 40\ \text{cm/s}$

Q27. Solution

Correct Answer: (B)

The rate of heat flow is given by

$$\frac{\Delta Q}{\Delta t}=-K\left(T-T_0\right) \Rightarrow \frac{\Delta Q}{\Delta t}=-K\left(T_{avg}-T_0\right)$$

$$\text{Initially } T_{avg}=\frac{T_1+T_2}{2}$$

Here, $T_1=98^{\circ}\mathrm{C}$

$T_2=86^{\circ}\mathrm{C}$

$$\text{(i) } \frac{\mathrm{ms} \times 12}{2}=-\mathrm{K}\left(\frac{98+86}{2}-22\right)$$

$$6=-\frac{\mathrm{K}}{\mathrm{ms}}\left[\frac{98+86}{2}-22\right] \quad (\because \Delta \mathrm{Q}=\mathrm{ms} \Delta \mathrm{T})$$

$6=-\frac{\mathrm{K}}{\mathrm{ms}}[70]$ ...(i)

Now, cool from $\mathrm{T}_1^{\prime}=75^{\circ}\mathrm{C}$

$\mathrm{T}_2=69^{\circ}\mathrm{C}$

$$\mathrm{T}_{\text{avg}}=\frac{\mathrm{T}_1^{\prime}+\mathrm{T}_2^{\prime}}{2}=\frac{75+69}{2}$$

(ii) $\frac{\mathrm{ms} \times 6}{\Delta \mathrm{t}}=-\mathrm{K}\left(\frac{75+69}{2}-22\right) \frac{6}{\Delta t}=-\frac{K}{\mathrm{ms}}(50)$ ...(ii)

Divide equation (ii) by (i), we have

$$\frac{6}{\Delta t(6)}=\frac{50}{70}$$

$$\Delta t=\frac{7}{5}=1.4 \text{ min}$$

Q28. Solution

Correct Answer: (C)

From the Gauss's law

$$\phi=\frac{\mathrm{q}_{\text{in}}}{\epsilon_0}=\frac{\mathrm{Q}}{\epsilon_0} \quad \because \mathrm{Q}=\mathrm{CV}$$

$$\therefore \phi=\frac{\mathrm{CV}}{\epsilon_0}$$

Q29. Solution

Correct Answer: (B)

Wavelength of monochromatic light,

$\lambda=600 \times 10^{-9}$ m

As fir first minima a $\sin \theta=\lambda$

$\Rightarrow \mathrm{a} \sin 30^{\circ}=600 \times 10^{-9}$

$\Rightarrow \mathrm{a}=1200 \times 10^{-9}$ m $=1.2 \mu \mathrm{m}$

Q30. Solution
Correct Answer: (B)

The relative velocity of a passenger with respect to source of sound (engine of train) is 0 . So there will be no Doppler's effect. So frequency heard is 400 Hz.

Q31. Solution
Correct Answer: (B)

In boundary surface diagram (1) the four lobes lie between y and z-axis $\left(d_{yz}\right)$ whereas, in boundary surface diagram (2) the four lobes lie on the x and y-axis $\left(d_{x^2-y^2}\right)$.

Q32. Solution
Correct Answer: (D)

6th period consists of 32 elements.

Q33. Solution
Correct Answer: (B)

$$r=0.529 \times \frac{n^2}{Z} \mathring{A}$$
$$r_3=0.529 \times \frac{3^2}{1} \Rightarrow r_4=0.529 \times \frac{4^2}{1}$$
$$\frac{r_4}{r_3}=\frac{4^2}{3^2}=\frac{16}{9} \Rightarrow r_4=\frac{16 r_3}{9}$$

Q34. Solution
Correct Answer: (A)

For first order reaction
$$k=\frac{1}{t} \ln \left(\frac{P_0}{P}\right) \Rightarrow \text{In}\left(\frac{P_0}{P}\right)=kt$$
$$\Rightarrow \ln \left(\frac{P}{P_0}\right)=-kt$$
On comaring with straight line equation $\mathrm{y}=\mathrm{mx}$
$$\mathrm{k}= \text{ slope } =3.465 \times 10^4$$
$$\mathrm{t}_{1/2}=\frac{\ln 2}{\mathrm{k}}=\frac{0.693}{3.465 \times 10^4}=2 \times 10^{-5} \mathrm{~s}$$

Q35. Solution
Correct Answer: (C)

Frenkel defect is due to dislocation of ion from its usual lattice site to interstitial position.

Q36. Solution

Correct Answer: (A)

When one mole of an ionic solid is formed from its constituent gaseous ions, the energy released is called the lattice energy.

Q37. Solution

Correct Answer: (A)

The specific conductance increases with concentration. The number of ions per cm^3 increase with increase of concentration.

Q38. Solution

Correct Answer: (C)

$\left(P+\frac{a}{V^2}\right)(V-b)=RT$; Here $\left(P+\frac{a}{V^2}\right)$ represents the intermolecular forces.

Q39. Solution

Correct Answer: (A)

Photochemical smog $\Rightarrow \mathrm{O}_3, \mathrm{PAN}, \mathrm{NO}_x$

Q40. Solution

Correct Answer: (A)

Colloid of liquid in liquid is called emulsion. Colloid of liquid in solid is gel.

Q41. Solution

Correct Answer: (B)

$\frac{1}{2}\mathrm{H}_2+\frac{1}{2}\mathrm{Cl}_2 \longrightarrow \mathrm{HCl}$

$\Delta H_{\mathrm{HCl}}=\sum \text{ B.E. of reactant } -\sum \text{ B.E. of products}$

$-90=\frac{1}{2}\times 430+\frac{1}{2}\times 240-\text{ B.E. of HCl}$

$\therefore \quad \text{B.E. of HCl}=215+120+90=425 \text{ kJ mol}^{-1}$

Q42. Solution

Correct Answer: (A)

Chromatography paper contains water trapped in it, which acts as the stationary phase.

Q43. Solution
Correct Answer: (D)

Vitamin B_1 - Thiamin Vitamin
B_6 - Pyridoxine

Q44. Solution
Correct Answer: (B)

High density polythene is formed when addition polymerisation of ethene takes place in a hydrocarbon solvent in presence of catalyst such as Ziegler-Natta catalyst.

Q45. Solution
Correct Answer: (A)

Mg burns in CO_2 to give MgO and C.

Q46. Solution
Correct Answer: (D)

Alkenes having double bonds with two different groups on each end of the double bond show geometrical isomerism. $A_2b_2c_2$, A_2b_2cd, A_2 bcde.

Q47. Solution
Correct Answer: (B)

$2BF_3 + 3LiBH_4 \rightarrow 2\ B_2H_6 + 3LiF$

Q48. Solution
Correct Answer: (A)

These drugs induce sleep and are habit forming common example of hypnotic drugs are luminal and saconal.

Q49. Solution
Correct Answer: (A)

1° amines (aliphatic and aromatic) react with $CHCl_3/KOH$ to yield isocyanide (foul smelling) This is known as carbylamine test which is not given by 2° and 3° amines.

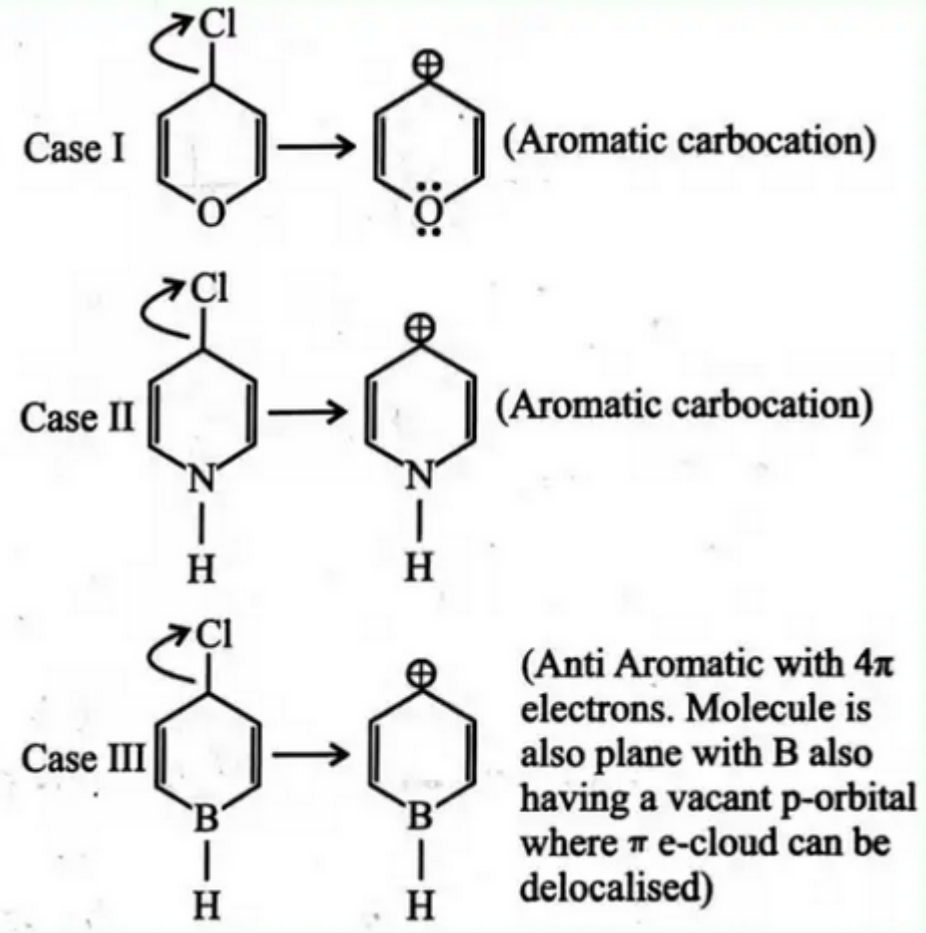
Q50. Solution
Correct Answer: (A)

Urea acts as stablising agent. It prevents the decomposition of H_2O_2.

Q51. Solution

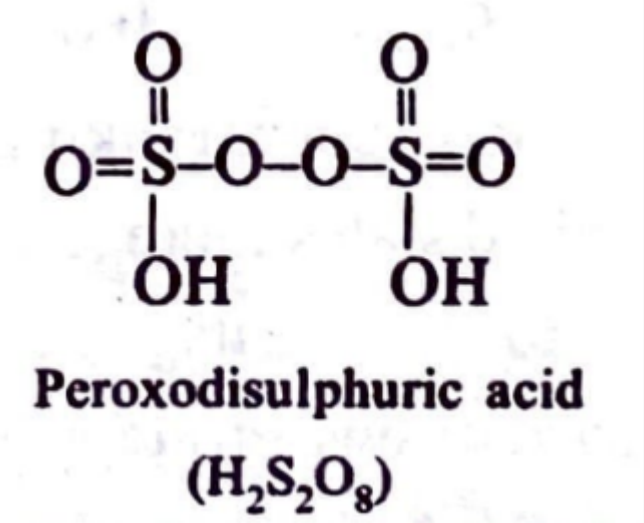
Correct Answer: (D)

Out of case I & case II, case Ill, carbocation is more stable because N shows more better +M (being less electro - ve) Stability of carbocation is the major concern here.



Q52. Solution

Correct Answer: (B)



Q53. Solution
Correct Answer: (A)

Pyrolusite (It is MnO_2)

Q54. Solution
Correct Answer: (C)

Reduction potential is maximum for $S_2O_8{}^{2-}$ therefore, it is the strongest oxidising agent amongst the given species.

Q55. Solution
Correct Answer: (D)

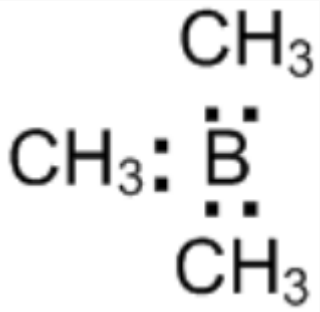
Phenol has active (acidic) hydrogen so it reacts with CH_3MgI to give CH_4, and not anisole
$C_6H_5OH + CH_3Mgl \rightarrow CH_4 + C_6H_5OMgl$

Q56. Solution
Correct Answer: (C)

$[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$ are ionisation isomers.

Q57. Solution
Correct Answer: (D)

According to Lewis concept "Acids are electron acceptor and bases are electron donor" Since, electron deficient compounds also have an ability to accept electrons, these are regarded as acids.
Trimethylborane $(CH)_3$ B have incomplete octet thus, act as Lewis acid.



Q58. Solution
Correct Answer: (D)

This reaction proceeds in the presence of anhydrous $AlCl_3$ or $CuCl$.

Q59. Solution
Correct Answer: (B)

$\Delta T_f = K_f \times m = 1.86 \times 0.5 = 0.93^\circ\text{C};$
$T_f = -0.93^\circ\text{C}$

Q60. Solution
Correct Answer: (A)

Gangue is the commercially worthless material which contaminates the ore.

Q61. Solution
Correct Answer: (A)

'Eloquent' means fluent or persuasive in speaking or writing.

Q62. Solution
Correct Answer: (A)

'Nefarious' means morally bad in principles or practice.

Q63. Solution
Correct Answer: (D)

If a person is very fat, you euphemistically call him or her corpulent. Emaciated, on the other hand, means extremely thin or weak because of illness or lack of food.

Q64. Solution
Correct Answer: (A)

The sentence has a general tone which is in simple past tense with the usages of verb+s because the subject is singular.

Q65. Solution
Correct Answer: (B)

The mention of 'now we live in Delhi' suggests that they lived in London in the past. So for that we use 'used to'.

Q66. Solution
Correct Answer: (A)

A school of psychology argues that motorcycling - like gambling or skydiving - is one of the manifestations of impulse control disorder, a condition in which an individual cannot resist the impulse or temptation to perform an act harmful for oneself or others.

Q67. Solution
Correct Answer: (D)

With six of its neighbours ranking high on global roster of failed states there is a renewed warning for India to reassess its policy towards them and safeguard its own strategic interests.

Q68. Solution
Correct Answer: (B)

Murali was a vendor of sweets.

Q69. Solution
Correct Answer: (A)

Murali's main customers were children.

Q70. Solution
Correct Answer: (D)

At the stroke of nine in the morning, Murali would stand in front of the school with his tray of sweets.

Q71. Solution
Correct Answer: (A)

$$\text{XXIV} \times 2$$

$$\Rightarrow 24 \times 2 = 48$$

Similarly,

$$\text{XIV} \times 2 = 14 \times 2 = 28$$

Q72. Solution
Correct Answer: (C)

Answer figure (c) is the best fit to the question figure.

Q73. Solution

Correct Answer: (D)

do re me → he is late

fa me la → she is early

so ti do → he leaves soon

Hence re → Late

Q74. Solution

Correct Answer: (A)

P(+) ⇔ S (–)

J – N(+) B(+) – C(–)

The gender of J is unknown, therefore J may be brother or sister of C.

Q75. Solution

Correct Answer: (D)

Given Series is:

22, 45, 91, 183, 367

×2 + 1 ×2 + 1 ×2 + 1 ×2 + 1

Q76. Solution

Correct Answer: (D)

$$\left(\frac{3}{2}-1\right)=\frac{1}{2} \Rightarrow 1^{\text{st}} \text{ row}$$
$$\left(\frac{8}{3}-2\right)=\frac{2}{3} \Rightarrow 2^{\text{nd}} \text{ row}$$
$$\left(\frac{19}{5}-3\right)=\frac{4}{5} \Rightarrow 3^{\text{rd}} \text{ row}$$

Q77. Solution

Correct Answer: (C)

Meaningful order of words:

2. Vegetable

↓

6. Cut

↓

4. Prepare

↓

3. Package

↓

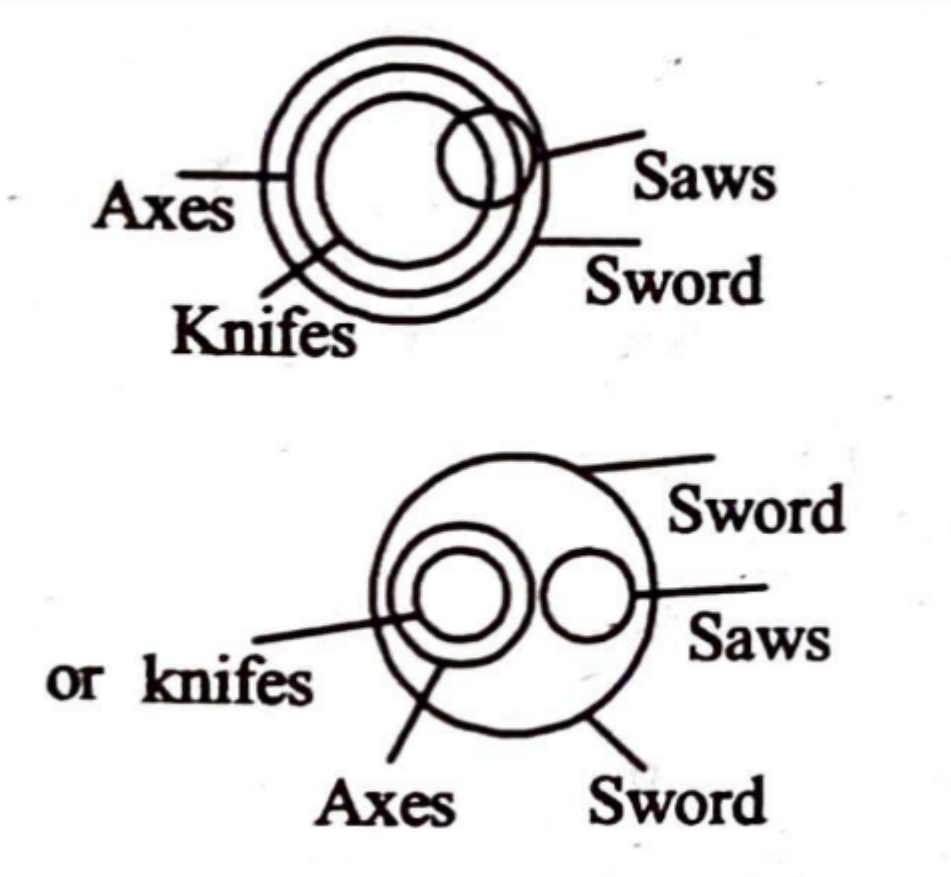
5. Store

↓

1. Serve

Q78. Solution

Correct Answer: (A)



Hence, either 'some axes are saws' or 'no axes are saws

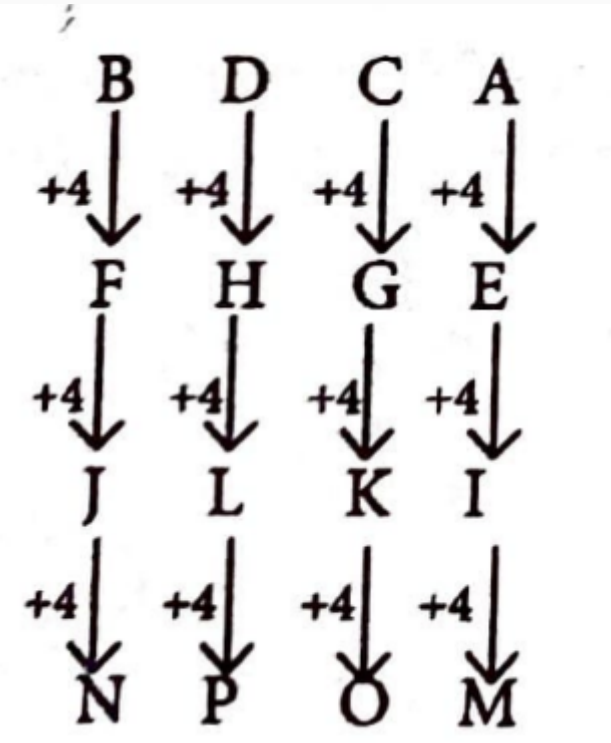
Q79. Solution

Correct Answer: (D)

To determine which word can be formed using the letters of REMEMBERING, we check whether all the letters of the given options appear in REMEMBERING. NEGRO Letters: N, E, G, R, O "O" is not present in "REMEMBERING" → Cannot be formed ❌ AGREE Letters: A, G, R, E, E "A" is not present in "REMEMBERING" → Cannot be formed ❌ RAIN Letters: R, A, I, N "A" is not present in "REMEMBERING" → Cannot be formed ❌ GREEN ✅ Letters: G, R, E, E, N All these letters are present in "REMEMBERING" → Can be formed ✔ Answer: (4) GREEN ✅

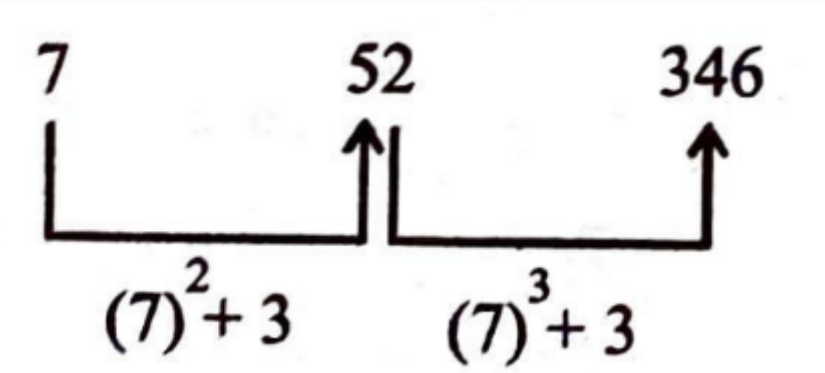
Q80. Solution

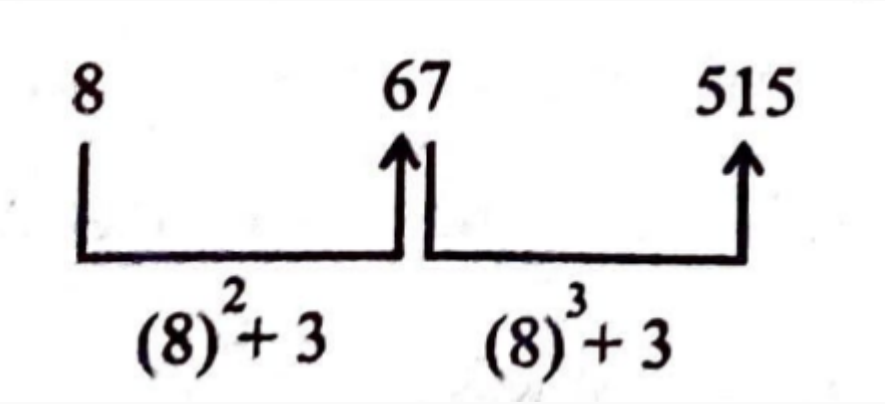
Correct Answer: (C)



Q81. Solution

Correct Answer: (A)

As, 

Similarly, 

Q82. Solution
Correct Answer: (D)

Lily, Daisy, Datura all have outer part white and inner part yellow.
Jasmine also has outer part white and inner part yellow.

Q83. Solution
Correct Answer: (C)

$142 : 15 \Rightarrow (1 \times 4 \times 2) + (1 + 4 + 2) = 15$
$234 :? \Rightarrow (2 \times 3 \times 4) + (2 + 3 + 4) = 33$

Q84. Solution
Correct Answer: (A)

$2678 \Rightarrow 2 \times 6 \times 7 \times 8 = 672$
$4325 \Rightarrow 4 \times 3 \times 2 \times 5 = 120$
$6931 \Rightarrow 6 \times 9 \times 3 \times 1 = 162$
$2132 \Rightarrow 2 \times 1 \times 3 \times 1 = 161$

Q85. Solution
Correct Answer: (A)

As, $(4 + 6 + 9) \times 2 = 38$
$(8 + 9 + 11) \times 2 = 56$
Similarly,
$(12 + 13 + 11) \times 2 = 72$

Q86. Solution
Correct Answer: (A)

From options
(a) $37, 4 \rightarrow 37 \times 4 = 148$ {odd one}
(b) $24, 7 \rightarrow 24 \times 7 = 168$
(c) $42, 4 \rightarrow 42 \times 4 = 168$
(d) $14, 12 \rightarrow 14 \times 12 = 168$

Q87. Solution

Correct Answer: (C)

$\div \rightarrow >$	$\times \rightarrow +$
$+ \rightarrow \div$	$- \rightarrow =$
$> \rightarrow \times$	$= \rightarrow <$
$< \rightarrow -$	

$5 > 2 \times 1 - 3 > 4 < 1$

Using the proper notations in (c), we get the statement as:

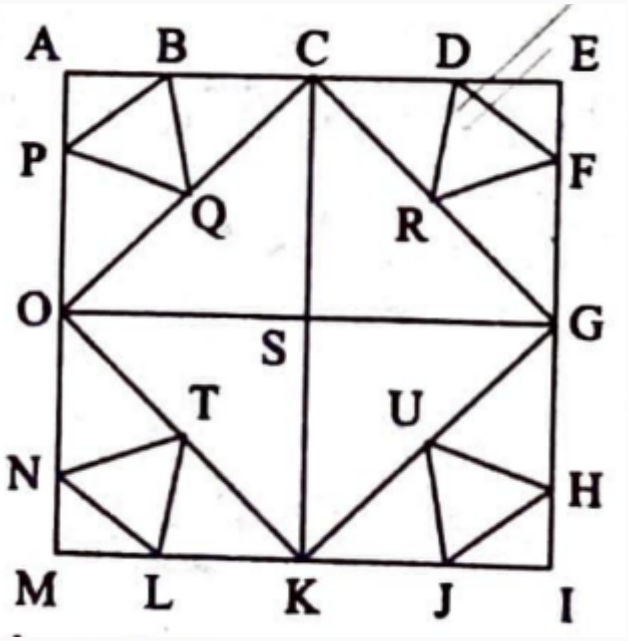
$5 \times 2 + 1 = 3 \times 4 - 1$

$10 + 1 = 12 - 1$

$11 = 11$

Q88. Solution

Correct Answer: (C)

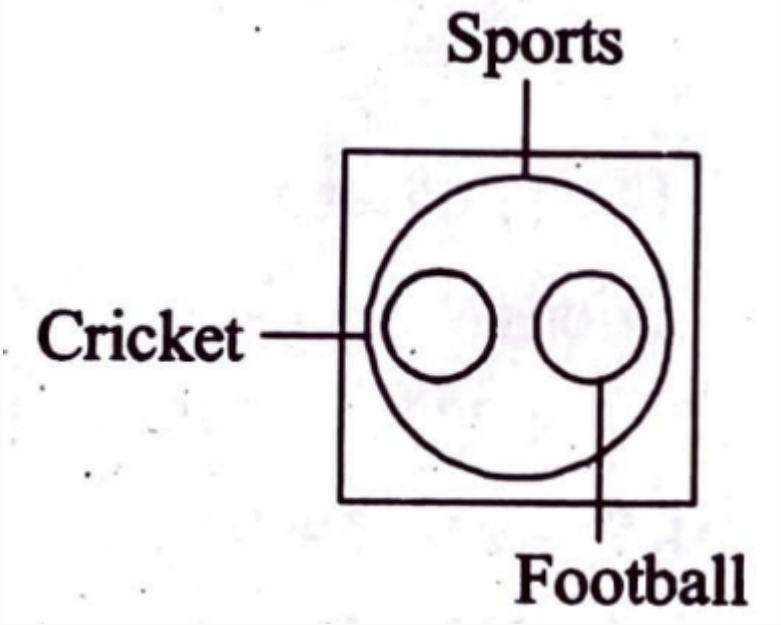


The triangles are:
△ABP; △BQP; △PQO; △BCQ
△ACO; ΔCSO; ΔCDR; ΔDRF;
△DEF; △RFG; △CSG; △ECG;
ΔIJH; ΔJUH; ΔGHU; ΔJKU
ΔGIH; ΔSGK; ΔONT; ΔNTL;
△NML; ΔTLK; ΔMOK; ΔSOK;
ΔCGO; ΔGKC; ΔKGO; ΔCOK

Q89. Solution

Correct Answer: (D)

The Venn diagrams best represent the relationship between - Sports, Cricket, Football figures are shown below.



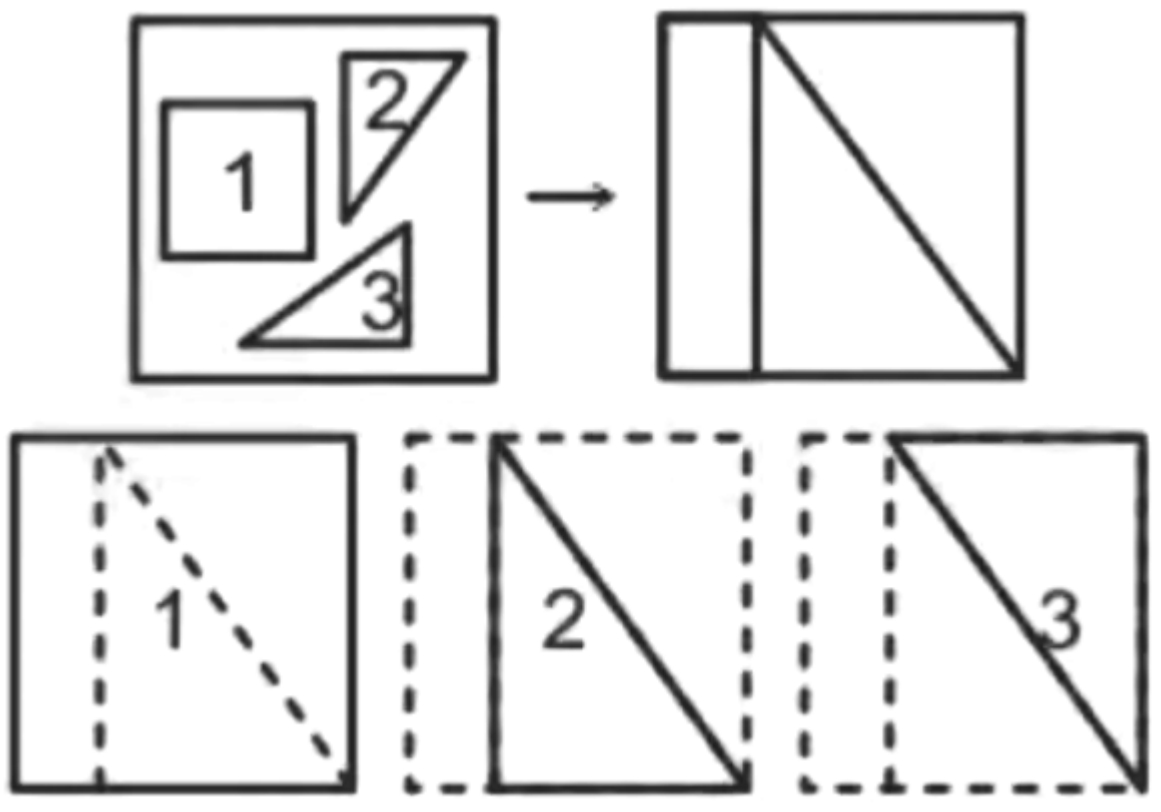
Both Cricket and Football are comes under the category of sport but the cricket and football are different types of sports so they are separated here.

Hence, correct answer is option (d).

Q90. Solution

Correct Answer: (B)

As shown in the following figure we can number the question figure can find the answer figure as follows,



Hence, "option figure A" is the correct answer.

Q91. Solution

Correct Answer: (B)

Z is 7 minimum at $\left(\frac{3}{2}, \frac{1}{2}\right)$

Q92. Solution

Correct Answer: (C)

Given boolen expression are

$\sim p \vee q \equiv p \rightarrow q$ and $\sim q \wedge p \equiv \sim (p \rightarrow q)$

Negation of $\sim (p \rightarrow q) \rightarrow (p \rightarrow q)$

is $\sim (p \rightarrow q) \wedge (\sim (p \rightarrow q))$ i.e., $\sim (p \rightarrow q)$

Q93. Solution

Correct Answer: (A)

A vector perpendicular to the plane is

$(\hat{i}-2\hat{j}-\hat{k})\times(3\hat{i}-2\hat{j}-\hat{k})$

$$= \begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & -1 \\ 3 & -2 & -1 \end{matrix} = -2\hat{j}+4\hat{k}$$

$$\Rightarrow \text{ unit vector } \hat{a} = \frac{-2\hat{j}+4\hat{k}}{\sqrt{4+16}} = \frac{-2\hat{j}+4\hat{k}}{2\sqrt{5}}$$

Angle between the unit vector and $\vec{r}=\hat{i}+\hat{j}+\hat{k}$

$= \cos^{-1}\frac{\vec{r}\cdot\hat{a}}{|\vec{r}|\cdot|\hat{a}|} = \cos^{-1}\frac{1}{\sqrt{15}} = \tan^{-1}\sqrt{14}$

Q94. Solution

Correct Answer: (B)

Given curve is $x=3t^2+1$

$$\ldots(i)$$

$$\therefore \frac{dx}{dt} = 6t$$

Second curve is $y=t^3-1$

$$\ldots(ii)$$

$$\therefore \frac{dy}{dt} = 3t^2$$

$\therefore \frac{dy}{dx} = \frac{dy}{dt}\times\frac{dt}{dx} = 3t^2\times\frac{1}{6t} = \frac{t}{2}$

But from (i) when $x=1$

we have $1=3t^2+1 \Rightarrow 3t^2=0 \Rightarrow t=0$

$\therefore$ When x $=1$ then t $=0$ $\quad \therefore \frac{\mathrm{dy}}{\mathrm{dx}} = \mathbf{0}$

Hence, slope of the tangent to the curve $=0$

Q95. Solution

Correct Answer: (A)

We have $\tan 5\theta = \cot 2\theta$

$\Rightarrow \tan 5\theta = \tan\left(\frac{\pi}{2} - 2\theta\right) \ldots$

$\left[\because \tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta\right]$

$\Rightarrow 5\theta = n\pi + \frac{\pi}{2} - 2\theta \Rightarrow 7\theta = n\pi + \frac{\pi}{2}$

$\Rightarrow \quad \theta = \frac{n\pi}{7} + \frac{\pi}{14}.$

Q96. Solution

Correct Answer: (C)

X – X – X – X – X. The four digits 3, 3, 5,5 can be arranged at $(-)$ places in $\frac{4!}{2!2!} = 6$ ways. The five digits $2, 2, 8, 8, 8$ can be arranged at (X) places in $\frac{5!}{2!3!}$ ways $= 10$ ways

Total no. of arrangements $= 6 \times 10 = 60$ ways

Q97. Solution

Correct Answer: (B)

Ratio $= -\left(\frac{3}{-2}\right) = \frac{3}{2}$

$\therefore$ Required co-ordinates of the points are

$\left[\frac{6-6}{5}, \frac{10+3}{5}, \frac{-14+24}{5}\right] = \left(0, \frac{13}{5}, 2\right).$

$\therefore a + b - c = 0 + 5 - 2 = 3$

Q98. Solution

Correct Answer: (A)

When the two lines intersect then shortes distance between them is zero i.e.

$$\frac{\left(\vec{a}_2 - \vec{a}_1\right) \cdot \vec{b}_1 \times \vec{b}_2}{\vec{b}_1 \times \vec{b}_2} = 0$$

$\Rightarrow \left(\vec{a}_2 - \vec{a}_1\right) \cdot \vec{b}_1 \times \vec{b}_2 = 0$

where $\vec{a}_1 = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b}_1 = k\hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{a}_2 = 2\hat{i} + 3\hat{j} + \hat{k}, \hat{b}_2 = 3\hat{i} + k\hat{j} + 2\hat{k}$

$$\Rightarrow \begin{matrix} 1 & 1 & -2 \\ k & 2 & 3 \\ 3 & k & 2 \end{matrix} = 0$$

$\Rightarrow \quad 1(4 - 3k) - 1(2k - 9) - 2\left(k^2 - 6\right) = 0$

$\Rightarrow \quad -2k^2 - 5k + 25 = 0 \Rightarrow k = -5 \text{ or } \frac{5}{2}$

Q99. Solution

Correct Answer: (D)

Since, $f(x)=\begin{cases} x^2-1, & 0<x<2 \\ 2x+3, & 2\le x<3 \end{cases}$

Now, $\lim_{x\to 2^-} f(x)=\lim_{x\to 2^-}\left(x^2-1\right)$

$=\lim\limits_{h\to 0}\left[(2-h)^2-1\right]=\lim\limits_{h\to 0}\left[4+h^2-4h-1\right]$

$=\lim\limits_{h\to 0}\left[h^2-4h+3\right]=0-0+3=3$

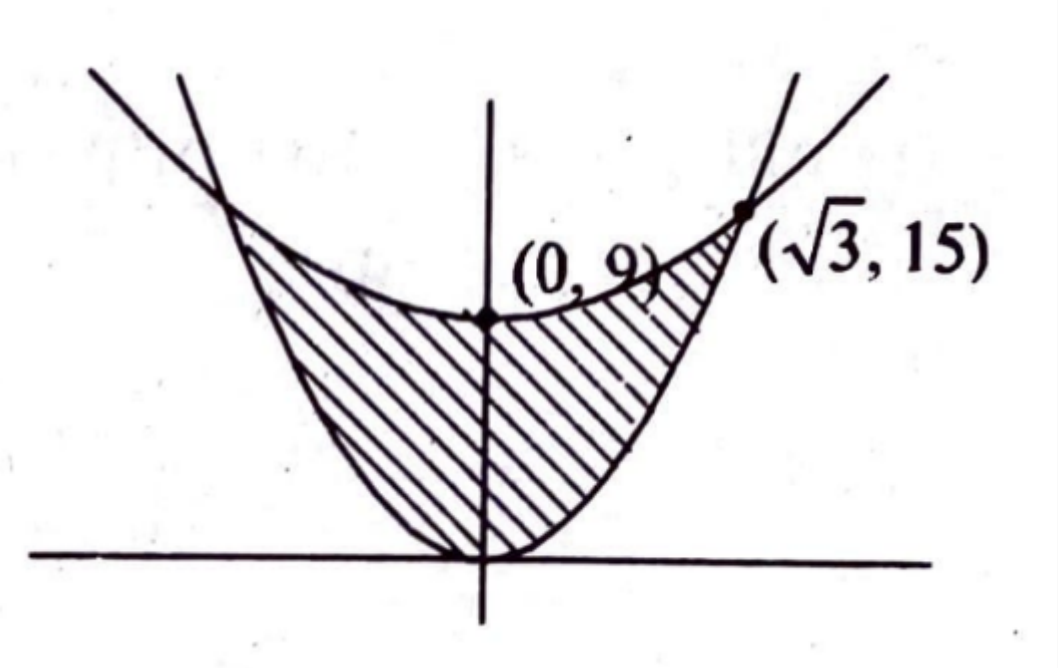
and $\lim_{x\to 2^+} f(x)=\lim_{x\to 2^+}(2x+3)$

$=\lim_{h\to 0}[2(2+h)+3]=\lim_{h\to 0}[4+2h+3]=7$

Hence the quadratic equation whose roots are 3 and 7 is given by $x^2-(3+7)x+(3+7)=0 \Rightarrow x^2$ $-10x+21=0$

Q100. Solution

Correct Answer: (B)



$$\text{Required area } =2\int_0^{\sqrt{3}}\left(2x^2+9-5x^2\right)dx$$

$=2\left[9x-x^3\right]_0^{\sqrt{3}}=2[9\sqrt{3}-3\sqrt{3}]=12\sqrt{3}.$

Q101. Solution

Correct Answer: (D)

$\mathrm{n(A)}=1000, \mathrm{n(B)}=500, \mathrm{n(A\cap B)}\ge 1, n(A\cup B)=p; n(A\cup B)=n(A)+n(B)-n(A\cap B)$

$\mathrm{p}=1000+500-\mathrm{n(A\cap B)}\Rightarrow 1\le \mathrm{n(A\cap B)}\le 500$

Hence $p\le 1499$ and $p\ge 1000\Rightarrow 1000\le p\le 1499$

Q102. Solution

Correct Answer: (C)

Let $P(n)=2 \cdot 4^{2n+1}+3^{3n+1}$

Then $P(1)=2.4^3+3^4=209$, which is divisible by 11 but not divisible by 2,7 or 27 .

Further, let $P(k)=2.4^{2k+1}+3^{3k+1}$ is divisible by 11 , that is,

$2.4^{2k+1}+3^{3k+1}=11q$ for some integer q. Now

$$\begin{aligned}
\mathrm{P}(\mathrm{k}+1) &= 2 \cdot 4^{2\mathrm{k}+3}+3^{3\mathrm{k}+4} \\
&= 2 \cdot 4^{2\mathrm{k}+1} \cdot 4^2+3^{3\mathrm{k}+1} \cdot 3^3 \\
&= 16 \cdot 2 \cdot 4^{2\mathrm{k}+1}+27 \cdot 3^{3\mathrm{k}+1} \\
&= 16 \cdot 2 \cdot 4^{2\mathrm{k}+1}+(16+11) \cdot 3^{3\mathrm{k}+1} \\
&= 16\left[2 \cdot 4^{2\mathrm{k}+1}+3^{3\mathrm{k}+1}\right]+11 \cdot 3^{3\mathrm{k}+1} \\
&= 16 \cdot 11\mathrm{q}+11 \cdot 3^{3\mathrm{k}+1} \\
&= 11\left(16\mathrm{q}+3^{3\mathrm{k}+1}\right)=11 \mathrm{~m}
\end{aligned}$$

where $m=16q+3^{3k+1}$ is another integer.

$\therefore \mathrm{P}(\mathrm{k}+1)$ is divisible by 11 .

Q103. Solution

Correct Answer: (C)

Given equation is

$(p-q)x^2+(q-r)x+(r-p)=0$

By using formula for finding the roots viz: $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$, we get

$$x=\frac{(r-q) \pm \sqrt{(q-r)^2-4(r-p)(p-q)}}{2(p-q)}$$

$$\Rightarrow x=\frac{(r-q) \pm (q+r-2p)}{2(p-q)}=\frac{r-p}{p-q}, 1$$

Q104. Solution

Correct Answer: (A)

We have, $I=\int 1 \cdot \tan^{-1} \sqrt{\mathrm{x}} \mathrm{dx}$

Using by parts,

$$\begin{aligned}
I &= \tan^{-1} \sqrt{x} \cdot (x)-\int \frac{1}{1+x} \times \frac{1}{2\sqrt{x}} \times x dx \\
&= x \tan^{-1} \sqrt{x} \cdot -\int \frac{x}{(1+x) 2\sqrt{x}} dx \\
&= x \tan^{-1} \sqrt{x}-\int\left(\frac{1+x}{(1+x) 2\sqrt{x}}-\frac{1}{(1+x) 2\sqrt{x}}\right) dx \\
&= x \tan^{-1} \sqrt{x}-\int \frac{dx}{2\sqrt{x}}+\int \frac{dx}{2\sqrt{x}(1+x)} \\
&= x \tan^{-1} \sqrt{x}-\sqrt{x}+\tan^{-1} \sqrt{x}+C \\
&= (x+1) \tan^{-1} \sqrt{x}-\sqrt{x}+C
\end{aligned}$$

Q105. Solution

Correct Answer: (C)

One vertex of square is $(-4, 5)$ and equation of one diagonal is $7x - y + 8 = 0$ Diagonal of a square are perpendicular and bisect each other

Let the equation of the other diagonal be $y = mx + c$ where m is the slope of the line and c is the y-intercept.

Since this line passes through $(-4, 5)$

$\therefore \quad 5 = -4m + c$ ...(i)

Since this line is at right angle to the line

$7x - y + 8 = 0$ or $y = 7x + 8$, having slope $= 7$,

$\therefore 7 \times \mathrm{m} = -1$ or $\mathrm{m} = \frac{-1}{7}$

Putting this value of m in equation (i) we get

$5 = -4 \times \left(\frac{-1}{7}\right) + c$

or $5 = \frac{4}{7} + c \quad$ or $\quad c = 5 - \frac{4}{7} = \frac{31}{7}$

Hence equation of the other diagonal is

$y = -\frac{1}{7}x + \frac{31}{7}$ or $7y = -x + 31$ or $x + 7y - 31 = 0$ or $x + 7y = 31$

Q106. Solution

Correct Answer: (A)

Given $f(x) = \log(1 + x) - \frac{2x}{2+x}$

$$f'(x) = \frac{1}{1+x} - \frac{(2+x)(2) - 2x}{(2+x)^2}$$

$$= \frac{1}{1+x} - \frac{4}{(2+x)^2}$$

$$= \frac{(2+x)^2 - 4 - 4x}{(1+x)(2+x)^2} = \frac{x^2}{(1+x)(2+x)^2} > 0 \text{ for all}$$

$x \in (0, \infty)$

Thus, given function f(x) is increasing on $(0, \infty)$.

Q107. Solution

Correct Answer: (D)

For $f(x)$ to be defined, we must have

$x - \sqrt{1 - x^2} \geq 0$ or $x \geq \sqrt{1 - x^2} > 0$ or $x^2 \geq 1 - x^2$

or $x^2 \geq \frac{1}{2}$.

Now, $x^2 \geq \frac{1}{2} \Rightarrow \left(x - \frac{1}{\sqrt{2}}\right)\left(x + \frac{1}{\sqrt{2}}\right) \geq 0$

$\Rightarrow x \leq -\frac{1}{\sqrt{2}} \quad$ or $\quad x \geq \frac{1}{\sqrt{2}}$

Also, $1 - x^2 \geq 0$ or $x^2 \leq 1$.

Also, $x^2 \leq 1 \Rightarrow (x - 1)(x + 1) \leq 0 \Rightarrow -1 \leq x \leq 1$

Thus, $x > 0, x^2 \geq \frac{1}{2}$ and $x^2 \leq 1 \Rightarrow x \in \left[\frac{1}{\sqrt{2}}, 1\right]$

Q108. Solution

Correct Answer: (A)

We have

$S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \ldots\ldots\infty$ ...(i)

Multiplying both sides by $\frac{1}{3}$, we get $\frac{1}{3}S = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \frac{10}{3^4} + \ldots\ldots\ldots\infty$ ...(ii)

Subtracting eqn. (ii) from eqn. (i), we get

$$\frac{2}{3}S = 1 + \frac{1}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} + \ldots\ldots\ldots\infty$$

$$\Rightarrow \frac{2}{3}S = \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} + \ldots\ldots\ldots\infty$$

$$\Rightarrow \frac{2}{3}S = \frac{\frac{4}{3}}{1 - \frac{1}{3}} = \frac{4}{3} \times \frac{3}{2} \Rightarrow S = 3$$

Q109. Solution

Correct Answer: (A)

Given $f(x) = \frac{4x+3}{6x-4}$

Let $y = \frac{4x+3}{6x-4}$,

$$\Rightarrow 6xy - 4y = 4x + 3 \Rightarrow x(6y - 4) = 3 + 4y$$

$$\Rightarrow x = \frac{3 + 4y}{6y - 4}$$

$$f^{-1}(x) = \frac{3 + 4x}{6x - 4}$$

Q110. Solution

Correct Answer: (A)

Given hyperbola is

$9x^2 - 16y^2 - 36x + 96y - 252 = 0$

$\Rightarrow 9\left(x^2 - 4x\right) - 16\left(y^2 - 6y\right) = 252$

$\Rightarrow 9\left(x^2 - 4x + 4\right) - 16\left(y^2 - 6y + 9\right)$

$= 252 + 36 - 144$

$\Rightarrow 9(x-2)^2 - 16(y-3)^2 = 144$

$\Rightarrow \frac{(x-2)^2}{16} - \frac{(y-3)^2}{9} = 1$ ...(i)

Put $x = X + 2$ and $y = Y + 3$

$\therefore$ Equation (i) becomes $\frac{X^2}{16} - \frac{Y^2}{9} = 1$

Now, vertices are $X = \pm a$ wehre $a = 4$ and $Y = 0$

Hence, vertices are $(6, 3), (-2, 3)$.

Q111. Solution

Correct Answer: (D)

Here, $y^x = e^{y-x}$

Taking log on both sides, we get

$\log y^x = \log e^{y-x}$

$\left(\because \log a^b = b \log a \text{ and } \log e = 1\right)$

$\Rightarrow x \log y = (y - x) \log e \Rightarrow x \log y = y - x \ldots (i)$

On differentiating w.r.t. x, we get

$\frac{d}{dx}(x \log y) = \frac{d}{dx}(y - x)$

(using product rule)

$$\Rightarrow x\left(\frac{1}{y}\right)\frac{dy}{dx} + \log y(1) = \frac{dy}{dx} - 1$$

$$\Rightarrow \frac{dy}{dx}\left(\frac{x}{y} - 1\right) = -1 - \log y$$

$$\Rightarrow \frac{dy}{dx}\left[\frac{y}{(1 + \log y)y} - 1\right] = -(1 + \log y)$$

$$\left[\because \text{ from eq. } (i) \cdot x = \frac{y}{(1 + \log y)}\right]$$

$$\Rightarrow \frac{dy}{dx}\left[\frac{1 - 1 - \log y}{1 + \log y}\right] = -(1 + \log y)$$

$\Rightarrow \frac{dy}{dx} = -\frac{(1+\log y)^2}{-\log y}$

$\Rightarrow \frac{dy}{dx} = \frac{(1+\log y)^2}{\log y}$

Q112. Solution

Correct Answer: (C)

$$AB = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$$

$$AB = \begin{bmatrix} abc - abc & b^2c - b^2c & bc^2 - bc^2 \\ -a^2c + a^2c & -abc + abc & -ac + ac \\ a^2b - a^2b & ab^2 - ab^2 & abc - abc \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

Q113. Solution

Correct Answer: (B)

Let (h, k) be the coordinates of the foot of the perpendicular from the point $(2, 3)$ on the i $x + y - 11 = 0$.
Then, the slope of the perpendicular line is $\frac{k-3}{h-2}$. Again the slope of the given line $x + y - 11 = 0$ is -1
Using the condition of perpendicularity of lines, we have
$\left(\frac{k-3}{h-2}\right)(-1) = -1$ or $k - h = 1$ ...(i)
Since (h, k) lies on the given line, we have, $h + k - 11 = 0$ or $h + k = 11$ ...(ii)
Solving (i) and (ii), we get $h = 5$ and $k = 6$. Thus $(5, 6)$ are the required coordinates of the foot of the perpendicular.

Q114. Solution

Correct Answer: (C)

For mutually exclusive events
$\mathrm{P(A \cup B) = P(A) + P(B) \Rightarrow P(A) = \frac{2}{7}}$

Q115. Solution

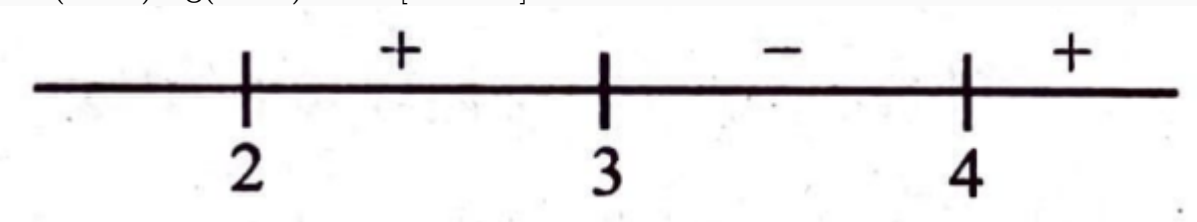
Correct Answer: (D)

The sample space is $\mathrm{S = \{HH, HT, TH, TT\}}$
Let E be the event of getting atleast one tail
$\therefore \mathrm{E = \{HT, TH, TT\}}$
$\therefore$ Required probability p
$$= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} = \frac{\mathrm{n(E)}}{\mathrm{n(S)}} = \frac{3}{4}$$

Q116. Solution

Correct Answer: (B)

Clearly $x > 2$. Write the given inequality as
$\left(x^2 - 6x + 8\right) \log(x - 2) > 0 \Leftrightarrow (x - 2)(x - 4) \log(x - 2) > 0$
$\Leftrightarrow (x - 4) \log(x - 2) > 0 \quad [\therefore x > 2]$



Q117. Solution

Correct Answer: (B)

We have, $f(x)=\sqrt{1+\cos^2\left(x^2\right)}$ ...(i)

On differentiating (i) w.r.t.x, we get

$f'(x)=\frac{-2\sin x^2\cos x^2}{\sqrt{1+\cos^2 x^2}}(x) \Rightarrow f'(x)=\frac{-\sin 2x^2}{\sqrt{1+\cos^2 x^2}}(x)$ ...(ii)

Put, $x=\frac{\sqrt{\pi}}{2}$ in (ii), we get

$$\mathrm{f}'\left(\frac{\sqrt{\pi}}{2}\right)=-\frac{\sqrt{\pi}}{2}\cdot\frac{\sin 2\left(\frac{\pi}{4}\right)}{\sqrt{1+\frac{1}{2}}}$$

$$=-\frac{\sqrt{\pi}}{2}\cdot\frac{\sin\frac{\pi}{2}}{\sqrt{\frac{3}{2}}}=-\sqrt{\frac{\pi}{6}}$$

Q118. Solution

Correct Answer: (C)

(c) Since $x\frac{dy}{dx}-y=x^4-3x$

$$\therefore \frac{dy}{dx}-\frac{y}{x}=x^3-3$$

Hence $IF=e^{\int Pdx}=e^{-\int\frac{1}{x}dx}=e^{-\log x}=\frac{1}{x}$

Q119. Solution

Correct Answer: (A)

$P(E\mid F)+P(\bar{E}\mid F)$

$=\frac{P(E\cap F)+P(\bar{E}\cap F)}{P(F)}=\frac{P((E\cup\bar{E})\cap F)}{P(F)}=\frac{P(F)}{P(F)}=1$

Q120. Solution

Correct Answer: (B)

The lines are $\frac{x}{6}=\frac{y+2}{6}=\frac{z-1}{1}$
and $\frac{x+1}{12}=\frac{y}{6}=\frac{z}{-1}$
Here,
$\vec{a}_1=-2\hat{j}+\hat{k}, b_1+6\hat{i}+6\hat{j}+\hat{k}, \vec{a}_2=-\hat{i},$
$\vec{b}_2=12\hat{i}+6\hat{j}-\hat{k}$

$$\vec{b}_1\times\vec{b}_2=\begin{matrix}\hat{i} & \hat{j} & k\\ 6 & 6 & 1\\ 12 & 6 & -1\end{matrix}=-12\hat{i}+18\hat{j}-36\hat{k}$$

$$\text{Shortest distance }=\frac{\left(\vec{a}_2-\vec{a}_1\right)\cdot\left(\vec{b}_1-\vec{b}_2\right)}{\vec{b}_1\times\vec{b}_2}$$

$$=\frac{|(-\hat{i}+2\hat{j}-\hat{k})\cdot(-12i+18\hat{j}-36\hat{k})|}{\sqrt{(-12)^2+(18)^2+(-36)^2}}$$

$$=\frac{|+12+36+36|}{\sqrt{1764}}=\frac{84}{42}=2$$

Q121. Solution

Correct Answer: (A)

$$\cos\left[2\cos^{-1}x+\sin^{-1}x\right]$$
$$=\cos\left[\cos^{-1}x+\cos^{-1}x+\sin^{-1}x\right]$$
$$=\cos\left[\cos^{-1}x+\pi/2\right]=-\sin\left[\cos^{-1}x\right]$$
$$=-\sin\left[\sin^{-1}\sqrt{1-x^2}\right]=-\sqrt{1-x^2}$$
$$=-\sqrt{1-\left(\frac{1}{5}\right)^2}=-\sqrt{\frac{24}{25}}=-\frac{2\sqrt{6}}{5}$$

Q122. Solution

Correct Answer: (B)

Suppose, $I=\int_0^{\pi/2}\cos xe^{\sin x}dx$
Let $\sin x=t\Rightarrow\cos xdx=dt$
$x\to 0\Rightarrow t\to 0$
and $x\to\frac{\pi}{2}\Rightarrow t\to 1$
So $I=\int_0^1 e^t dt=\left[e^t\right]_0^1=e-1$

Q123. Solution

Correct Answer: (C)

In the expansion of $\left(x^4+\frac{1}{x^3}\right)^{15}$, let T_r is the rth term

$$T_r={}^{15}C_{r-1}\left(x^4\right)^{15-r+1}\left(\frac{1}{x^3}\right)^{r-1}$$

$$={}^{15}C_{r-1}x^{64-4r-3r+3}=15_{C_{r-1}}x^{67-7r}$$

x^{18} occurs in this term

$\Rightarrow 18=67-7r\Rightarrow 7r=49\Rightarrow r=7.$

Q124. Solution

Correct Answer: (C)

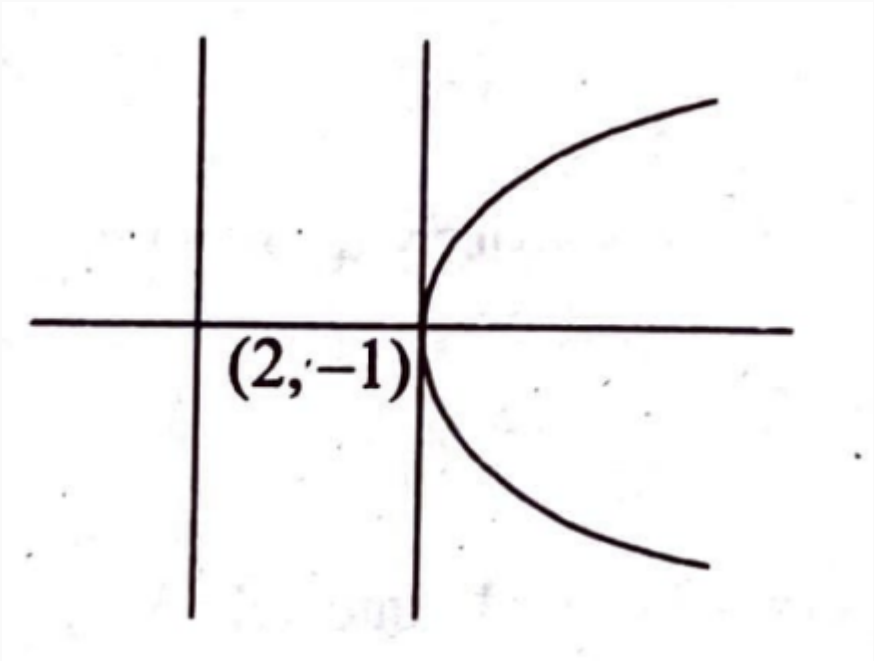
Here, $b^2-4ac=1^2-4\times\sqrt{5}\times\sqrt{5}=1-20=-19$

Therefore, the solutions are

$\frac{-1\pm\sqrt{-19}}{2\sqrt{5}}=\frac{-1\pm\sqrt{19i}}{2\sqrt{5}}$

Q125. Solution

Correct Answer: (B)



We are given that directrix of the parabola is $4x-3y=21$ and vertex of the parabola is $(2,-1)$

Now $a=\frac{|8+3-21|}{5}=\frac{10}{5}=2$

$\therefore$ latus rectum of the parabola $=4a=8$

Q126. Solution

Correct Answer: (D)

$$\frac{(1+i\sqrt{3})(2+2i)}{\sqrt{3}-i}=\frac{2+2\sqrt{3}i+2i-2\sqrt{3}}{\sqrt{3}-i}$$
$$=\frac{(2-2\sqrt{3})+(2\sqrt{3}+2)i}{\sqrt{3}-i}\times\frac{\sqrt{3}+i}{\sqrt{3}+i}$$
$$=\frac{2\sqrt{3}-6+2i-2\sqrt{3}i+6i+2\sqrt{3}i-2\sqrt{3}-2}{3+1}$$
$$=\frac{8i-8}{4}=-2+2i$$
$$\therefore \text{ Modulus } =\sqrt{(-2)^2+(2)^2}=2\sqrt{2}$$

Q127. Solution

Correct Answer: (C)

Here $h=0, k=2$ and $r=2$. Therefore, the required equation of the circle is :
$(x-0)^2+(y-2)^2=(2)^2$
or $x^2+y^2-4y+4=4$ or $x^2+y^2-4y=0$

Q128. Solution

Correct Answer: (D)

(a) Let $f(x)=\cos x$, then $f'(x)=-\sin x$.
In interval $\left(0, \frac{\pi}{2}\right), f'(x)<0$
Therefore, $f(x)$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$
(b) Let $f(x)=\cos 2x \Rightarrow f'(x)=-2\sin x 2x$
In interval $\left(0, \frac{\pi}{2}\right), f'(x)<0$
Because $\sin 2x$ will either lie in the first or second quadrant which will give a positive value.
Therefore, $f(x)$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$
(c) Let $f'(x)=\cos 3x$
$\Rightarrow \mathrm{f}'(\mathrm{x})=-3\sin 3\mathrm{x}$. In Interval $\left(0, \frac{\pi}{3}\right), \mathrm{f}'(\mathrm{x})<0$
Because $\sin 3\mathrm{x}$ will either lie in the first or second quadrant which will give a positive value.
Therefore, $\mathrm{f}(\mathrm{x})$ irstrictly decreasing on $\left(0, \frac{\pi}{3}\right)$.
When $\mathrm{x} \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$, then $\mathrm{f}'(\mathrm{x})>0$
Because $\sin 3x$ will lie in the third quadrant. Therefore, $f(x)$ is not strictly decreasing on
$\left(0, \frac{\pi}{2}\right)$
(d) Let $f(x)=\tan x \Rightarrow f'(x)=\sec^2 x$.
In Interval $\mathrm{x} \in \left(0, \frac{\pi}{2}\right), \mathrm{f}'(\mathrm{x})>0$
Therefore, $f(x)$ is not strictly decreasing on
$\left(0, \frac{\pi}{2}\right)$

Q129. Solution

Correct Answer: (A)

$$\text{Mean } (\bar{x}) = \frac{1+2+3+4+5}{5} = 3$$

$$\text{S.D } = \sigma = \sqrt{\frac{1}{5}(1+4+9+16+25) - 9} = \sqrt{11-9}$$

$$= \sqrt{2}$$

Q130. Solution

Correct Answer: (A)

Let $\mathrm{x} = \mathrm{a}\sin^2\theta$

then $dx = 2a\sin\theta\cos\theta d\theta$

$$\therefore \quad I = \int \frac{2a\sin\theta\cdot\cos\theta}{\sqrt{a\sin^2\theta\cdot a\cos^2\theta}} d\theta$$

$$= 2\int d\theta = 2\theta + c$$

$$= 2\sin^{-1}(\sqrt{x/a}) + c$$

Q131. Solution

Correct Answer: (A)

$c = f\lambda, \frac{1}{\lambda} = \frac{f}{c}$

$\frac{1}{\lambda} = R\left(\frac{1}{P^2} - \frac{1}{n^2}\right)$

$\Rightarrow f = Rc\left(\frac{1}{P^2} - \frac{1}{n^2}\right)$

$\Rightarrow f_2 = Rc\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = Rc\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5Rc}{36}$

$f_1 = Rc\left(\frac{1}{2^2}\right) = \frac{Rc}{4}$

$f_3 = Rc\left(\frac{1}{3^2}\right) = \frac{Rc}{9}$

$\therefore \quad f_1 - f_3 = Rc\left(\frac{1}{4} - \frac{1}{9}\right)$

$\therefore \quad f_1 - f_3 = f_2$

$\therefore \quad f_1 - f_2 = f_3$

Q132. Solution

Correct Answer: (C)

According to Kepler's law, $T^2 \propto a^3$.

Here, $a =$ semi-major axis $= \left(\frac{r_1+r_2}{2}\right)$.

$\therefore\ T^2 \propto \left(\frac{r_1+r_2}{2}\right)^3$

$T \propto \left(\frac{r_1+r_2}{2}\right)^{3/2} \propto (r_1+r_2)^{3/2}$.

Q133. Solution

Correct Answer: (C)

$\gamma_{mix} = 1 + \frac{2}{f_{mix}}$

$\frac{C_p}{C_v} = \frac{f_{mix}+2}{f_{mix}} = \frac{3}{2}$

$\Rightarrow \quad f_{mix} = 4$

$f_{mix} = \frac{n_1 f_1 + n f_2}{n_1 + n_2}$

$\Rightarrow 4 = \frac{2\times3+n\times5}{2+n}$

$\Rightarrow n = 2$ mole

Q134. Solution

Correct Answer: (D)

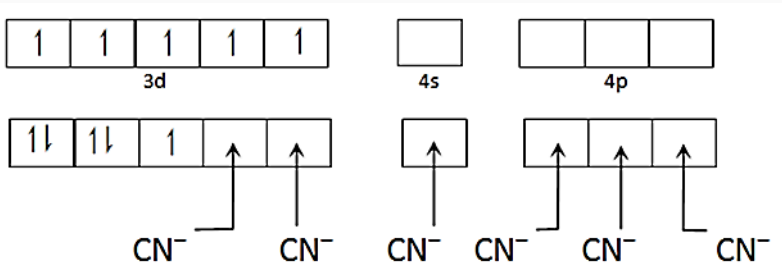
$K_3[Fe(CN)_6]$

Electronic configuration of $Fe = [Ar]4s^2 3d^6$

Electronic configuration of $Fe^{+3} = [Ar]3d^5$

Number of ligand (coordination numbr) $= 6$

Nature of ligand is strong field.



Hybridization of Fe is d^2sp^3 .

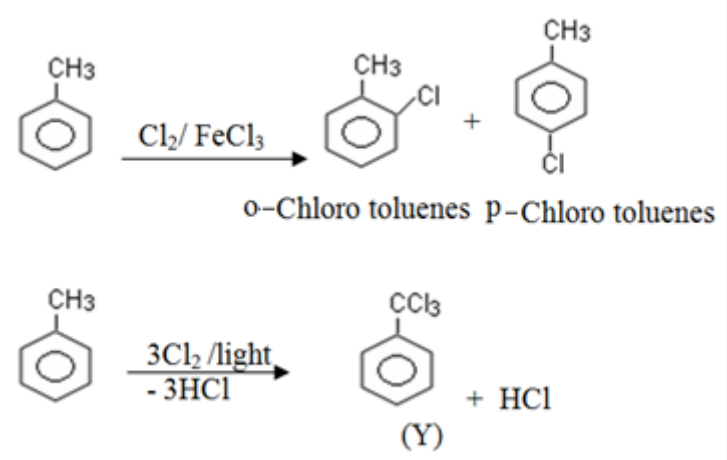
Q135. Solution

Correct Answer: (A)

$$\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{k_{a_1}}{k_{a_2}}}$$
$$\frac{1}{2} = \sqrt{\frac{2\times10^{-4}}{k_{a_2}}}$$
$$\frac{1}{4} = \frac{2\times10^{-4}}{k_{a_2}}$$
$$k_{a_2} = 8 \times 10^{-4}$$

Q136. Solution

Correct Answer: (D)



Q137. Solution

Correct Answer: (A)

According to the question,

Given information is, Given a series and we need to find the next term of given series,

Given series is, $6,\ 11,\ 24,\ \ 49,\ 90,\ ?$

So, the method used in this series I s, $+1^2+2^{2,},\ +2^2+3^2,\ +3^2+4^2,\ +4^2+5^2.....$

$11 = 6 + 1^2 + 2^2\ 24 = 11 + 2^2 + 3^2\ 49 = 24 + 3^2 + 4^2\ 90 = 49 + 4^2 + 5^2\ ? = 90 + 5^2 + 6^2 =$

Hence, the answer is 151.

Q138. Solution
Correct Answer: (B)

Given letter series,

B, F, J, P, ?

The pattern followed by the series is shown below:

B + 4 = F

F + 4 = J
J + 6 = P

On the basis of the above pattern, we can observe that
P + 6 = V

Q139. Solution
Correct Answer: (D)

We know that when you assist, you help someone. Save means your help is of a high degree. Similarly, when you request, you ask for help from someone and when you command, you are very strong in a request for what you ask.

Aid means to assist someone.

Oppose is to disagree with someone.
Decry is to depreciate something.

Hence, the correct answer is Command.

Q140. Solution
Correct Answer: (B)

Equation of normal at $P(6,3)$

$\frac{a^2x}{6}+\frac{b^2y}{3}=a^2+b^2$

It passes through $(9,0)$

$\frac{3}{2}a^2=a^2+b^2 \quad \Rightarrow \frac{3}{2}=\frac{a^2+b^2}{a^2}=1+\frac{b^2}{a^2}$

$\Rightarrow \; e=\sqrt{1+\frac{b^2}{a^2}}=\sqrt{\frac{3}{2}}.$

Q141. Solution

Correct Answer: (C)

$$\lim_{n\to\infty} \frac{1}{n^2}\sec^2\frac{1}{n^2} + \frac{2}{n^2}\sec^2\left(\frac{4}{n^2}\right) + ... + \frac{1}{n}\sec^2 1$$

$$= \lim_{n\to\infty} \frac{1}{n^2}\sec^2\frac{1}{n^2} + \frac{2}{n^2}\sec^2\left(\frac{4}{n^2}\right) + ... + \frac{n}{n^2}\sec^2\left(\frac{n^2}{n^2}\right)$$

$$= \lim_{n\to\infty}\sum_{r=1}^{r=n}\left(\frac{r}{n^2}\right)\sec^2\left(\frac{r}{n}\right)^2 = \lim_{n\to\infty}\sum_{r=0}^{r=n}\frac{1}{n}\left(\frac{r}{n}\right)\sec^2\left(\frac{r}{n}\right)^2$$

$$= \int_0^1 x\,\sec^2\left(x^2\right)dx = \frac{1}{2}\tan 1$$

Q142. Solution

Correct Answer: (A)

We have,

$$A = \begin{bmatrix} x & y & 2 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$$

$|A| = x(4-3) - y(2-3) + 2(1-2)$

$\Rightarrow |A| = x + y - 2$

And,

$$\text{adj}\left(\text{adj}\left(\text{adj}\left(\text{adj } A\right)\right)\right) = 3^{32} \cdot 5^{16}$$

Now,

$|\text{adj } A| = |A|^{n-1}$

$\Rightarrow |\text{adj } (\text{adj } A)| = |A|^{(n-1)^2}$

$\Rightarrow |(\text{adj } (\text{adj } (\text{adj } (\text{adj } A))))| = |A|^{(n-1)^4}$

Here, $n = 3$, therefore

$|(\text{adj } (\text{adj } (\text{adj } (\text{adj } A))))| = |A|^{(3-1)^4}$

$\Rightarrow |(\text{adj } (\text{adj } (\text{adj } (\text{adj } A))))| = |A|^{16} = 3^{32} \cdot 5^{16}$

$\Rightarrow |A|^{16} = 3^{16+16} \cdot 5^{16}$

$\Rightarrow |A|^{16} = 3^{16} \cdot 3^{16} \cdot 5^{16}$

$\Rightarrow |A|^{16} = (3 \cdot 3 \cdot 5)^{16}$

$\Rightarrow |A| = 45$

$\Rightarrow x + y - 2 = 45$

$\Rightarrow x + y = 47$

Number of matrix is equal to the number of positive integral solution of $x + y = 47$ i.e.

${}^{47-1}C_{2-1} = {}^{46}C_1 = 46$